Phase 1 (MVP) Subtask Deepening — epic → task → subtask (closes refinement R5)

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Revision: 2 **Last modified:** 2026-07-04T12:00:00Z **Rev 2:** Independent gap-analysis pass — spot-checked subtask acceptance criteria against `07-phase1-mvp-wbs.md` parent tasks (E01/E02/E03 sampled in depth); all carry falsifiable, captured-evidence acceptance and a §11.4.169 test-type set. No placeholder/TBD text found. Confirmed this document is what closes R5 for Phase 1 (the parent WBS itself stays at epic→task; the third tier lives here, as designed).

> Volume 7 (Phase Execution), document 3 of 5. This spec **closes refinement item R5** (`REFINEMENT_NOTES.md`): Phase 0 (`06-…`) formalizes a full epic→task→subtask 3-tier breakdown, but Phase 1 (`07-phase1-mvp-wbs.md`) stops at epic→task. Here every Phase-1 task `HVPN-P1-NNN` is decomposed into PR-sized subtasks `HVPN-P1-NNN.k` (§11.4.93/.54 ids), each carrying a concrete **acceptance criterion** (falsifiable, captured-evidence per §11.4.5/.69/.107), the **§11.4.169 test types** that gate it, and an **estimated complexity** (XS/S/M/L T-shirt, sizing-only `TARGET` — never a date, §11.4.6). It is **spec-only**: it describes the subtasks and how "done" is proven; it does not build them. Every subtask traces to its parent task's Desc/Deliverable/ Acceptance in `07-…`; nothing is invented beyond decomposing the stated work. These `.k` rows feed the `workable-items` DB (`workable-items-model.md` §7 docs_chain

> context). Companion: `dependency-graph.md` (the DAG these subtasks inherit), `subtask-deepening-p2.md`, `subtask-deepening-p3.md`.

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0. Deepening conventions

Each parent task gets a subtask table: `id` (`HVPN-P1-NNN.k`, monotonic per parent, §11.4.54) · **Subtask** (≥6-word title, §11.4.91) · **Acceptance** (falsifiable, captured-evidence) · **Tests** (§11.4.169 codes — `UNIT INT E2E FA SEC CHAOS STRESS PERF BENCH SCALE UI UX REC CHAL`, the `07-…` §1 vocabulary) · **Cx** (complexity XS≈0.5d, S≈1–2d, M≈2–3d, L≈3–5d — sizing `TARGET`).

A subtask is `complete` only when its test types are green with a `test_diary` evidence path (`workable-items-model.md` §9). Subtask complexity sums to ≈ the parent task's `07-…` Effort; it is a sizing aid, not a commitment (§11.4.6, `07-…` §22). Phase-0 entry-gate items (`HVPN-P1-001..006`) are prerequisites tracked in `06-…`; they are not re-deepened here.

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1. E01 — Foundation (repo, proto, codegen)

HVPN-P1-010 — Monorepo + submodule layout (`07-…` §6; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Scaffold the snake_case component tree (`helix-core/ helix-edge/ helix-go/ helix-proto/ helix-ui/ shims/ deploy/`)	each dir present + decoupled (§11.4.28/.29); `tree` capture matches [SYNTHESIS §6]	UNIT,FA	S
`.2`	Wire `.gitmodules` + per-repo `helix-deps.yaml` (§11.4.31) for the six incorporated submodules	`git submodule status` lists containers/ helix_qa/ challenges/ docs_chain/ security/ vision_engine	UNIT	XS
`.3`	`upstreams/` recipes + `install_upstreams` per repo (§11.4.36)	`git remote -v | grep -c push` = expected count on a fresh clone	FA	XS
`.4`	Assert **no active CI** (§11.4.156): workflows disabled/renamed	`CM-NO-ACTIVE-CI` green; `git ls-files` finds no `.github/workflows/*.yml`	FA,SEC	XS

HVPN-P1-011 — `helix-proto` agent protobuf + buf codegen (`07-…` §6; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Author `agent.proto` (`Coordinator`: Enroll/ WatchNetworkMap/ AdvertisePrefixes/ ReportStatus) per [02 §4]	`buf lint` clean; service matches the [02 §4] contract	UNIT	S
`.2`	Wire `buf generate` → Go + Dart + Rust stubs	one source → three generated clients compile	UNIT,FA	S
`.3`	`buf breaking` gate on incompatible change	a removed field FAILs `buf breaking` (captured)	FA	XS
`.4`		grep finds zero hand-rolled	SEC	XS

id	Subtask	Acceptance	Tests	Cx
	Assert no hand-written client code exists (§8 [04_P1])	RPC client; generated-only		

HVPN-P1-012 — REST OpenAPI + Dart/TS client gen (`07-…` §6; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Define `openapi/helix.v1.yaml` (enroll-tokens/devices/policies/networks/stream)	OpenAPI 3.1 validates	UNIT	S
`.2`	Generate Dart + TS clients; round-trip a stub server	generated client round-trips; spec↔handler drift FAILs the gen check	UNIT,FA	S

HVPN-P1-013 — Local dev infra via `containers` submodule (`07-…` §6; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`internal/testinfra` wrapper booting PG+Redis via `pkg/boot/pkg/compose/pkg/health` (§11.4.76/.161 rootless)	integration tests boot infra on demand; no ad-hoc docker	INT,FA	S
`.2`	`make test-infra-up/down` with orphan-free teardown (§11.4.14)	teardown leaves zero orphan containers (captured `podman ps`)	INT,FA	XS

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2. E02 — Store, RLS, no-log lint

Risk-first (§11.4.132): the irreversible correctness floor — ships before any feature reads/writes data.

HVPN-P1-020 — Migrations, enums, DB roles (`07-…` §7; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`goose` DDL: 11 tenant tables + enums per [02 §2.2]	migrations apply+rollback clean on fresh PG; `\dt` matches spec	UNIT,INT	M

id	Subtask	Acceptance	Tests	Cx
.2	Three roles `helix_owner`/ `helix_app`(non-super,non-bypass)/ `helix_sys`	role grants verified; `helix_app` lacks rolsuper/rolbypassrls	INT,SEC	S
.3	Assert no connection/traffic table present (privacy floor)	`\dt` shows zero `connections|flows|traffic|packets` table	SEC	XS

HVPN-P1-021 — `sqlc` typed queries + `store.WithTenant` (`07-…` §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	`sqlc`-generated compile-time-checked queries	generated `Queries` compiles; `sqlc vet` clean	UNIT	S
.2	`WithTenant(ctx,tid,fn)` tx-scoped `SET LOCAL app.tenant_id` + `WithSystem`	a query forgetting `WHERE tenant_id` still returns only the active tenant's rows (INT)	UNIT,INT	M

HVPN-P1-022 — FORCE RLS + tenant_isolation policies (`07-…` §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	`ENABLE`+`FORCE RLS` + `tenant_isolation USING`/ `WITH CHECK` on all 10 tenant tables	every tenant table has the policy; `\d+` capture	UNIT,INT	M
.2	`store/rls_guard.go` startup abort if `helix_app` can bypass RLS	startup aborts under a super role (captured exit)	UNIT,SEC	S
.3	Cross-tenant denial proof	a crafted tenant-A→tenant-B read returns ZERO rows (captured)	SEC	S
.4	Mid-tx drop rollback (no leak)	connection drop mid-tx rolls back; no partial cross-tenant write	CHAOS	S

HVPN-P1-023 — No-logging CI schema-lint + mutation (`07-…` §7; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`tools/schemalint` rejecting connection-log-shaped tables [02 §2.4]	green on the real schema; pre-build gate wired	UNIT,SEC	S
`.2`	Paired §1.1 mutation: inject `connections(src,dst,bytes,ts)` → lint FAILs	the planted table FAILs the lint (captured)	SEC	XS

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3. E03 — Identity, enrollment, PKI

HVPN-P1-030 — Dual identity: OIDC + anonymous device tokens (`07-…` §8; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	OIDC managed mode (Keycloak/ Authentik) → tenant/role mapping	an OIDC user maps to a tenant/role (INT)	UNIT,INT	M
`.2`	Anonymous device enroll-tokens (no PII, Mullvad posture)	an anonymous device enrolls with zero PII persisted (SEC asserts DB/ logs)	UNIT,SEC	S
`.3`	Token mint/store-hashed/consume-atomic	single-use, short-lived; concurrent double-consume rejected (SEC)	SEC,STRESS	S

HVPN-P1-031 — `Enroll` RPC (device key never leaves) (`07-…` §8; M · SLO2)

id	Subtask	Acceptance	Tests	Cx
`.1`	Token verify → IPAM-allocate → issue cert → persist device row (public WG key only)	device row holds only the WG public key (SEC asserts no private key)	UNIT,INT,SEC	M
`.2`			INT,E2E	S

id	Subtask	Acceptance	Tests	Cx
	Emit `device.enrolled`; first NetworkMap path	event well-formed; map issued		
.3	SLO2 timing: enroll → first NetworkMap < 2 s	captured histogram p99 < 2 s	E2E,PERF	S

HVPN-P1-032 — PKI: device-cert issuance + tenant CA (`07-…` §8; M)

id	Subtask	Acceptance	Tests	Cx
.1	`pki.Issue/Rotate/Revoke` short-lived (24 h) mTLS certs vs per-tenant CA	a cert authenticates `WatchNetworkMap` (INT)	UNIT,INT	M
.2	CA-root secrecy (KMS/offline) + secret-leak audit (§11.4.10.A)	CA private key never in any app log (audit green)	SEC	S

HVPN-P1-033 — Cert rotation + revoke-<1 s (`07-…` §8; M · SLO3, AC6)

id	Subtask	Acceptance	Tests	Cx
.1	Auto-rotate before expiry over the existing channel	rotation at the boundary works (captured)	UNIT,INT	S
.2	`device.revoke` → drop WG peer from every map + edge enforce + cert revoked	revoked cert rejected immediately (SEC)	SEC	M
.3	SLO3 timing: revoke → edge enforcement < 1 s	captured revoke→enforcement timing p99 < 1 s	PERF	S

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4. E04 — IPAM

HVPN-P1-040 — ULA `/48` provisioning + `AllocOverlayIP` (`07-…` §9; M, D4)

id	Subtask	Acceptance	Tests	Cx
.1	Provision `fd7a:helix:<rand>::/48`		UNIT,INT	S

id	Subtask	Acceptance	Tests	Cx
	per tenant (`::1`=gateway) on tenant create	each tenant gets a distinct ULA /48 (INT)		
`.2`	`AllocOverlayIP` deterministic + concurrency-safe	1000 concurrent allocations: no duplicate/gap (STRESS, captured)	UNIT,STRESS	M
`.3`	Restart-stable re-hydration of `next_host`	allocation resumes after restart with no reuse	INT	S

HVPN-P1-041 — `4via6` for colliding IPv4 LANs (`07-…` §9; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Stable site-id allocator per connector site	each site gets a stable id (INT)	UNIT,INT	S
`.2`	`4via6` route synthesis → `Peer.allowed_ips`	one client reaches the same IPv4 CIDR via two connectors, disambiguated by site (E2E curl each)	E2E	M

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5. E05 — Event backbone

HVPN-P1-050 — `events.Bus` Redis Streams + envelope (`07-…` §10; M, D3)

id	Subtask	Acceptance	Tests	Cx
`.1`	`events.Bus` interface + Redis Streams `XADD/XReadGroup/XACK` + §5.2 envelope	at-least-once delivery; idempotent reaction; replay harmless (INT)	UNIT,INT	M
`.2`	Consumer-crash re-delivery on restart	crash mid-handle re-delivers (CHAOS, captured)	CHAOS	S

HVPN-P1-051 — DLQ sweeper (`XAUTOCLAIM`) + metric (`07-…` §10; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`XAUTOCLAIM` reclaim + dead-letter routing + `helix_events_dlq_total`	poisoned event lands in DLQ after N retries; metric increments (captured)	UNIT,INT,CHAOS	S

HVPN-P1-052 — Event taxonomy producers (`07-…` §10; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Typed constructors for the [04_P1 §5.3] taxonomy (device., *connector.*, route.conflict, policy.*, gateway.failover)	each producer emits a well-formed envelope on its trigger (INT capture)	UNIT,INT	S

6. E06 — Policy model & compiler

HVPN-P1-060 — Spec parse + group/host resolution (`07-…` §11; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Parse `policies.spec` jsonb (groups/hosts/acls/exitNodes)	valid spec parses; malformed rejected	UNIT	S
`.2`	Resolve `group:*`→device-ids + `host`→CIDRs vs `advertised_prefixes`	unknown group/host rejected; ambiguity emits `route.conflict.detected`	UNIT,INT	M
`.3`	Reject an ACL granting a revoked device	revoked-device grant rejected (SEC)	SEC	S

HVPN-P1-061 — Two-artifact compiler (AllowedIPs + verdict map) (`07-…` §11; L · AC2/AC3)

id	Subtask	Acceptance	Tests
`.1`	Pure default-deny `compile(tenant,spec)→{visible,allowedIPs,exitNodes}`	golden-table tests pass; deterministic	UNIT
`.2`	Port-level verdict map (nftables/eBPF granularity)	a contractor granted	UNIT,SE

id	Subtask	Acceptance	Tests
		`554,80` cannot reach port 22 (verdict-map SEC)	
`.3`	Determinism property: same spec ⇒ byte-identical output	property test green over N runs (§11.4.50)	UNIT
`.4`	Perf: compile 1k-device tenant < 100 ms	captured timing < 100 ms	PERF

HVPN-P1-062 — Dry-run fail-closed validation + atomic activate/rollback (`07-…` §11; M · AC5)

id	Subtask	Acceptance	Tests	Cx
`.1`	Dry-run compile rejects unknown ref / uncovered dst / revoked grant	a bad policy never reaches the active version (fail-closed, captured)	UNIT,SEC	M
`.2`	Atomic version bump + `policy.compiled` emit + instant rollback	rollback restores the prior compiled set atomically (INT)	INT	S

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7. E07 — Coordinator & WatchNetworkMap

Critical path; the < 1 s convergence promise lives here.

HVPN-P1-070 — In-memory per-tenant topology graph (`07-…` §12; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Hydrate per-tenant graph from PG on boot (no durable coordinator tables)	graph reconstructable from PG + events alone (INT)	UNIT,INT	M
`.2`	Event-driven incremental subgraph mutation	applying an event mutates only the affected subgraph (INT)	INT,SCALE	M

HVPN-P1-071 — `buildMap(node)` per-node map (`07-…` §12; M · AC2/AC3)

id	Subtask	Acceptance	Tests	Cx
`.1`	`buildMap` → `NetworkMap{self,gateway,peers,dns,transport}` with `PeersVisibleTo` filter	a node never appears in another's map unless policy grants it (SEC, captured)	UNIT,SEC	M
`.2`	`relayEndpoint` via gateway (MVP hub-and-spoke)	reachability matches the Phase-0 slice	UNIT	S

HVPN-P1-072 — `WatchNetworkMap` stream loop (`07-…` §12; L · AC5)

id	Subtask	Acceptance	Tests	Cx
`.1`	mTLS-auth → snapshot/resume-from-`known_version` → delta+keepalive loop	reconnect with `known_version=N` resumes from deltas, no full resync (captured)	UNIT,INT,E2E	L
`.2`	Bounded per-stream send queue (slow-consumer shed)	a stalled consumer is shed without blocking others (CHAOS)	CHAOS	M

HVPN-P1-073 — Minimal-affected-set fan-out + reconcile metric (`07-…` §12; M · AC5/SLO1)

id	Subtask	Acceptance	Tests	Cx
`.1`	Compute minimal affected open-stream set; per-stream deltas (no full resync)	a 1-device change touches only the affected streams (INT count assertion)	UNIT,INT	M
`.2`	`helix_reconcile_seconds` histogram: event → delta-on-wire	SLO1 p99 < 1 s (captured histogram)	PERF	S

HVPN-P1-074 — 24 h soak / 10k-stream memory bound (`07-…` §12; M · SLO4)

id	Subtask	Acceptance	Tests	Cx
`.1`	10k simulated agents holding streams while flapping policies	SLO1 maintained under load (captured)	SCALE,STRESS	M
`.2`	24 h RSS time-series, bounded no-leak	coordinator memory bounded, no growth over 24 h (SLO4 graph)	PERF,SCALE	M

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8. E08 — API surface + authz

HVPN-P1-080 — Connect handlers on Gin + mTLS resolver (`07-…` §13; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	One multiplexed server: Connect (HTTP/2) alongside Gin REST	native agent + gRPC-Web caller both reach `Coordinator`	UNIT,INT	M
`.2`	`authDevice(ctx)` resolves identity from mTLS cert	an unauthenticated agent RPC is rejected (SEC)	SEC	S

HVPN-P1-081 — REST routes + OpenAPI + RBAC (`07-…` §13; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Gin routes (enroll-tokens/devices/ policies/networks/ audit)	routes match OpenAPI (INT)	UNIT,INT	M
`.2`	OIDC/session/API-token + RBAC (admin/operator/ member)	a `member` cannot mint enroll-tokens (403, SEC)	SEC	S
`.3`	RLS backstop (defense-in-depth)	RLS blocks cross-tenant read even if RBAC bypassed (INT)	INT,SEC	S

HVPN-P1-082 — `GET /v1/stream` WS/SSE fan-out (`07-…` §13; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Stream control-plane	a policy apply surfaces "policy v N active" to a	UNIT,INT,E2E	S

id	Subtask	Acceptance	Tests	Cx
	events to Console over WS/SSE	connected Console within the convergence window (E2E)		

HVPN-P1-083 — `AdvertisePrefixes` + `ReportStatus` RPCs (`07-…` §13; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`AdvertisePrefixes` accepts connector CIDRs → `connector.prefixes.changed`	advertised prefixes flow into the coordinator graph (INT)	UNIT,INT	S
`.2`	`ReportStatus` carries presence/transport/rtt only (no bytes/flows)	SEC asserts the message has no byte/ flow/ destination field	SEC	S

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9. E09 — helix-core (Rust client/ connector)

The shared client/connector/edge crate; durable evolution of the Phase-0 interfaces.

HVPN-P1-090 — `Transport` trait + plain-UDP + MASQUE (`07-…` §14; L · AC4, D1)

id	Subtask	Acceptance	Tests	Cx
`.1`	Harden the trait (`send/recv/kind/effective_mtu`, unreliable datagrams) into `helix-transport`	same trait drives client/ connector/ edge byte-for-byte	UNIT	M
`.2`	Production `plain_udp` impl	throughput baseline captured (BENCH)	UNIT,BENCH	S
`.3`	Production `masque` (quinn+h3+hand-rolled CONNECT-UDP/HTTP-Datagram,	MASQUE flow classifies HTTP/3, no WG signature (tshark, SEC)	UNIT,SEC,E2E	L

id	Subtask	Acceptance	Tests	Cx
	RFC 9298/9297/9221)			
`.4`	MASQUE goodput vs plain ≥ 50%	captured BENCH ≥ 50% of plain	BENCH	S

HVPN-P1-091 — `helix-wg` boringtun wrapper (`07-…` §14; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Wrap `Tunn` (handshake/ encrypt/decrypt/4 verdicts)	in-process handshake + data round-trip identical plaintext	UNIT	M
`.2`	Sustained 1 GB transfer + 2-min rekey	no handshake stall (STRESS); rekey at boundary works (captured)	E2E,STRESS	M

HVPN-P1-092 — Orchestrator + network-map reconciler (`07-…` §14; L · AC5)

id	Subtask	Acceptance	Tests	Cx
`.1`	Three-loop orchestrator (tun→wg→transport, transport→wg→tun, timer)	loops drive a plain-UDP slice (E2E)	UNIT,E2E	M
`.2`	Reconciler consuming streamed `MapUpdate` (snapshot/delta) without restart	a delta adds/ removes a peer with no reconnect of unrelated state (FA)	FA	M
`.3`	Mid-stream coordinator-restart re-sync from `known_version`	re-syncs cleanly (CHAOS)	CHAOS	S

HVPN-P1-093 — Auto-escalation ladder (`07-…` §14; M · AC4)

id	Subtask	Acceptance	Tests	Cx
`.1`	Walk `TransportPolicy.order` (plain-udp→lwo→masque-h3) on handshake failure, honor `allow_user_override`	with plain WG blocked, client escalates to MASQUE and stays up (E2E `transport=="masque-h3"`)	UNIT,E2E,SEC	M

id	Subtask	Acceptance	Tests	Cx
.2	Deterministic ladder across N runs (§11.4.50)	identical escalation outcome over N=3	UNIT	S

HVPN-P1-094 — Kill-switch + DNS-leak protection (`07-…` §14; L · AC7)

id	Subtask	Acceptance	Tests	Cx
.1	Core state machine drives platform firewall on Down/ Reconnecting	host-side pcap during a forced drop shows ZERO leaked packets (SEC)	UNIT,SEC	L
.2	DNS pinned to overlay resolver while up; no plaintext DNS on drop	zero plaintext DNS leaked (SEC capture)	SEC	M
.3	State survives an abrupt interface flap	kill-switch holds across flap (CHAOS)	CHAOS	S

HVPN-P1-095 — `helix-ffi` (flutter_rust_bridge v2) surface (`07-…` §14; M)

id	Subtask	Acceptance	Tests	Cx
.1	`start(ClientConfig)/stop()` + codegen	Dart drives connect/ disconnect; codegen clean	UNIT,FA	S
.2	`status_stream(StreamSink<TunnelStatus>)` mirroring the broadcast channel	Dart renders live status from the stream alone (UI golden + FA ×3)	UNIT,UI,FA	M

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10. E10 — helix-edge (data plane)

Language fixed by Phase-0 G4 (default lean Rust, shares `helix-transport`).

HVPN-P1-100 — MASQUE termination on `:443/udp` + masquerade (`07-…` §15; L · AC4)

id	Subtask	Acceptance	Tests	Cx
.1	Terminate MASQUE/ H3, extract HTTP Datagrams → WG → kernel WG	a valid client tunnels through (E2E, captured)	UNIT,E2E	L
.2	Native decoy site for non-CONNECT-UDP probes	a scanner at `:443` sees a plain website (SEC)	SEC	M
.3	CPU/Gbps within the G4 budget	captured BENCH within budget	BENCH	S

HVPN-P1-101 — Kernel-WG peer programming + relay (`07-…` §15; M · AC2)

id	Subtask	Acceptance	Tests	Cx
.1	Program kernel WG peers from the coordinator map	peers appear in `wg show` (INT)	UNIT,INT	M
.2	Hub-and-spoke relay peer↔peer via gateway	client reaches an authorized connector LAN host through the relay (E2E curl)	E2E	M

HVPN-P1-102 — Edge verdict-map enforcement + revoke drop (`07-…` §15; M · AC3/AC6, SLO3)

id	Subtask	Acceptance	Tests	Cx
.1	Apply the compiler's port-level verdict map (nftables/eBPF)	an unauthorized port is dropped (AC3 SEC, captured)	UNIT,INT,SEC	M
.2	Drop a revoked device's sessions on `device.revoked`	revoke removes the kernel peer < 1 s (SLO3, captured); negligible p99 add (PERF)	PERF,SEC	S

HVPN-P1-103 — Edge hardening posture (`07-…` §15; S)

id	Subtask	Acceptance	Tests	Cx
.1	Rootless Podman, read-only rootfs, seccomp, `NET_ADMIN`-only, no SSH	container runs unprivileged except `NET_ADMIN` (SEC)	SEC	S
.2	Fail-static: killing control plane doesn't	established tunnels survive a control-	CHAOS	S

id	Subtask	Acceptance	Tests	Cx
	drop data-plane tunnels	plane kill (CHAOS capture)		

11. E11 — helix-ui (Flutter)

All UI under OpenDesign tokens (§11.4.162) — light+dark, no ad-hoc CSS.

HVPN-P1-110 — `helix_design` system (OpenDesign tokens, Material 3) (`07-…` §16; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Brand-tokenized palette + signature components (ConnectButton/ StatusChip/ExitPicker/ ShieldIndicator/ AdaptiveScaffold), light+dark	visual-regression goldens pass for every component in both themes (UI)	UNIT,UI	M
`.2`	No overlapping/overlaid labels (§11.4.162 layout invariant)	layout-audit golden clean; palette reacts to connection state (REC vision-verified §11.4.159)	UI,REC	S

HVPN-P1-111 — Shared shell `runHelixApp(flavor,…)` + Riverpod binding (`07-…` §16; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	One tree → three flavors (Access/ Connector/Console)	same widget tree renders all three flavors (UI)	UNIT,UI	M
`.2`	Riverpod providers = pure function of the FFI status stream	UI state deterministic over a replayed status stream (FA)	FA	S

HVPN-P1-112 — Access app screens (enroll → connect → reach) (`07-…` §16; L · AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	Enroll (paste/scan token) → connect → ExitPicker/ authorized-network list	golden tests for each screen (UI)	UI	M

id	Subtask	Acceptance	Tests	Cx
`.2`	Real enroll→connect→reach-LAN journey window-scoped MP4 (§11.4.143/.159)	vision verdict confirms the authorized host loaded (UX/REC)	UX,E2E,REC	M

HVPN-P1-113 — Console (web) admin screens (`07-…` §16; L · AC5/AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	Devices/enroll-tokens/policy editor (validate→activate→rollback) + live `/v1/stream` feed	a policy edit reflects on devices < 1 s with a live "policy v N active" feed (E2E/REC, AC5)	UI,E2E,REC	L
`.2`	RBAC hides admin actions from `member`	member sees no admin action (SEC)	SEC	S

HVPN-P1-114 — Connector daemon UI/headless control (`07-…` §16; M · AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	Enroll + advertise CIDRs + show prefixes/ reachability; headless daemon mode	a connector advertises a LAN; its prefixes appear in Console + a client's map (E2E)	UI,E2E,REC	M

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12. E12 — Platform tunnel shims

MVP platforms only: iOS, Android, Linux. The shim is the only platform-specific code.

HVPN-P1-120 — Linux tun shim (Rust-native) (`07-…` §17; S · AC1/AC2)

id	Subtask	Acceptance	Tests	Cx
`.1`	Linux core drives the TUN	the Access Linux build runs the full	E2E,FA	S

id	Subtask	Acceptance	Tests	Cx
	directly (reference for the netns rig)	enroll→connect→reach journey on the netns rig (E2E)		

HVPN-P1-121 — Android `VpnService` + JNI shim (`07-…` §17; M · AC7/AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	`VpnService` owns the TUN; JNI loads `helix-core.so`	tunnel up on a real phone (device REC)	E2E,UX,REC	M
`.2`	Kill-switch → `VpnService` always-on/lockdown	kill-switch blocks plaintext on drop (SEC capture)	SEC	S

HVPN-P1-122 — iOS `NEPacketTunnelProvider` shim (within G3 budget) (`07-…` §17; L · AC7/AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	`NEPacketTunnelProvider` configures settings + pumps `packetFlow` ↔ Rust core	tunnel up on a real device (device REC)	E2E,UX,REC	L
`.2`	Honor the G3 memory ceiling over a sustained transfer	captured Instruments RSS within ceiling (PERF)	PERF	M
`.3`	Kill-switch + DNS-leak hold	zero plaintext/ DNS leak on drop (SEC)	SEC	S

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13. E13 — Deploy

HVPN-P1-130 — `helixvpnctl` bootstrap + ops CLI (Cobra) (`07-…` §18; M · AC1)

id	Subtask	Acceptance	Tests	Cx
`.1`	`init` generates tenant/ admin/CA/overlay-`/48`/ creds/gateway-keys/ quadlets	`helixvpnctl init` on a fresh host produces a runnable deploy (FA, captured)	UNIT,FA	M
`.2`	`gateway keys/enroll-token/policy apply/`	`policy apply ./policy.jsonc`	E2E	S

id	Subtask	Acceptance	Tests	Cx
	`device revoke` subcommands	validates + activates (E2E)		

HVPN-P1-131 — Podman quadlets (rootless, systemd-managed) (`07-…` §18; M · AC1)

id	Subtask	Acceptance	Tests	Cx
`.1`	Render `helixd/helix-edge/postgres/redis` quadlets, rootless, read-only rootfs	`systemctl --user start helixvpn-pod` brings the stack up rootless (FA)	INT,FA	M
`.2`	`helixd` connects to PG as the non-superuser role (RLS enforced)	the app role cannot bypass RLS (SEC)	SEC	S

HVPN-P1-132 — On-demand integration infra via `containers` submodule (`07-…` §18; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	QA harness boots+tears down the full stack via the submodule (§11.4.76)	the E2E suite boots+tears down the stack itself (FA, captured)	INT,FA	S

HVPN-P1-133 — Compose dev-loop + K8s Phase-2 shape (reference) (`07-…` §18; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Docker Compose dev-loop (local) + reference stateless-coordinator K8s manifest	dev compose boots PG+Redis+helixd; the K8s manifest is lint-valid (not deployed at MVP)	INT	S

14. E14 — QA, SLOs, DoD certification

Anti-bluff capstone: the §11.4.40 full-suite retest, §11.4.107 recorded evidence, §11.4.27 100%-type coverage.

HVPN-P1-140 — netns + DPI + netem E2E rig (CI-ready) (`07-…` §19; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Promote the Phase-0 rig fed by the real control		E2E,FA	M

id	Subtask	Acceptance	Tests	Cx
	plane (lanA + nft WG-block + tc netem)	`make e2e` runs the full slice unattended		
`.2`	Re-runnable `-count=3` identically (§11.4.98)	three identical runs (captured)	FA	S

HVPN-P1-141 — SLO instrumentation + dashboards (`07-…` §19; S · SLO1-4)

id	Subtask	Acceptance	Tests	Cx
`.1`	Prometheus metrics + Grafana-as-code + alert rules at the §2 SLO targets	all four SLOs measured + asserted in CI thresholds (captured histograms, alert-on-breach)	PERF,FA	S

HVPN-P1-142 — Security test suite (`07-…` §19; L · AC3/AC6/AC7/AC8)

id	Subtask	Acceptance	Tests	Cx
`.1`	RLS cross-tenant denial + key-never-leaves + mTLS-reject + kill-switch/DNS pcap + revoke timing + secret-leak audit (§11.4.10.A)	every assertion green with captured evidence	SEC,SCALE	L
`.2`	Paired §1.1 mutations (drop an RLS policy / weaken kill-switch) FAIL their gate	each mutation FAILs (captured)	SEC,CHAOS	M

HVPN-P1-143 — Stress + chaos suite (§11.4.85) (`07-…` §19; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Sustained load + concurrency + boundary across the control plane	no deadlock/leak under 10× concurrency	STRESS,SCALE	M
`.2`	Failure injection (Redis kill / PG drop / coordinator SIGKILL / disk-full)	every fault recovers to a consistent state with a captured `recovery_trace.log`	CHAOS	M

HVPN-P1-144 — HelixQA Challenge bank + vision-verified recordings (`07-…` §19; M · AC1-AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	`challenges/helix_qa` bank dispatching real user journeys, PASS only on captured evidence	the 8-AC journey set scores PASS with a vision-confirmed MP4 per AC (no frozen/bluff frame §11.4.107)	CHAL,REC,UX	M
`.2`	Analyzer self-validated (golden-good/bad); broken build FAILs	a deliberately broken build scores FAIL	CHAL	S

HVPN-P1-145 — Full-suite retest + DoD certification gate (`07-…` §19; M · AC1-AC9/SLO1-4)

id	Subtask	Acceptance	Tests	Cx
`.1`	§11.4.40 release-gate sweep on a clean baseline (pre/post-build + on-device + meta-mutation + Challenge + no-log lint + CONTINUATION sync)	all of §21 green simultaneously on a from-zero install (§11.4.108 fresh deploy)	FA,E2E,CHAL	M

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15. E15 — Governance & release

HVPN-P1-150 — Workable-items DB projection + docs-chain sync (`07-…` §20; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`cmd/workable-items/` loader: parse leaves → upsert `items/test_diary` (git-tracked DB §11.4.95)	md↔db byte-identical round-trip (§11.4.93)	UNIT,INT,FA	M
`.2`	`.docs_chain/contexts/wbs.yaml` registered (§11.4.106)	a leaf edit re-syncs exports out-	INT,FA	S

id	Subtask	Acceptance	Tests	Cx
		of-the-box; `verify` is the CI gate		

HVPN-P1-151 — Release tagging + multi-upstream publish
(`07-…` §20; S · AC1-AC9)

id	Subtask	Acceptance	Tests	Cx
`.1`	Cut `<HELIX_RELEASE_PREFIX>-1.0.0-mvp` (§11.4.151) on main + every owned submodule	`git tag -l '<PREFIX>-*'` enumerates the whole release across repos	FA	S
`.2`	Publish to all upstreams via merge-onto-latest-main (no force-push §11.4.113)	the tag is reachable on every remote; HVPN-P1-145 cert report is the tag's evidence	FA	S

16. Subtask roll-up + coverage note

Epic	Parent tasks	Deepened subtasks	Notes
E01 Foundation	4	12	repo/proto/codegen/dev-infra
E02 Store+RLS+lint	4	11	risk-first correctness floor
E03 Identity/PKI	4	11	revoke-<1s + key-never-leaves
E04 IPAM	2	5	ULA /48 + 4via6 (D4)
E05 Events	3	4	Redis Streams (D3)
E06 Policy compiler	3	9	two-artifact, default-deny
E07 Coordinator	5	11	<1s convergence + 24h soak
E08 API+authz	4	8	Connect+REST+RBAC+RLS backstop
E09 helix-core	6	16	Transport/WG/orchestrator/kill-switch/FFI
E10 helix-edge	4	9	MASQUE term + verdict-map + fail-static

Epic	Parent tasks	Deepened subtasks	Notes
E11 helix-ui	5	10	OpenDesign + 3 flavors
E12 shims	3	6	Linux/Android/iOS
E13 deploy	4	7	helixvpnctl + quadlets
E14 QA/DoD	6	11	rig/SLO/security/chaos/ Challenge/cert
E15 governance	2	4	items-DB + release tag
Total	**59**	**134**	—

Subtask complexity sums to ≈ the parent task effort (`07-…` §22, ~273 person-days excl. Phase-0 prereqs); it is a **sizing aid, not a commitment** (§11.4.6). Every subtask is anti-bluff: its acceptance is a captured-evidence assertion (§11.4.5/.69/.107), metadata-only PASS is forbidden (§11.4.1), and it is `complete` only with a `test_diary` evidence path (`workable-items-model.md` §9). The `.k` ids are monotonic per parent and never renumbered (§11.4.54). The full set of these subtasks feeds the §11.4.93 DB via the `subtask-deepening-p1.md` docs_chain source (`workable-items-model.md` §7).

Sources verified

- `07-phase1-mvp-wbs.md` §6–§20 (every task Desc/Deliverable/ Acceptance/Effort/Tests), §21 traceability, §22 effort roll-up — read 2026-06-26.
- `06-phase0-spike-wbs.md` §0.1 (the 3-tier breakdown R5 aligns to) — read 2026-06-26.
- `REFINEMENT_NOTES.md` R5 (subtask-tier asymmetry to close) — read 2026-06-26.
- Sibling `workable-items-model.md` (§3 field mapping, §6 test diary, §9 complete-definition), `dependency-graph.md` (the inherited DAG) — authored this volume.
- Constitution anchors §11.4.5/.6/.40/.54/.58/.69/.85/.91/.93/.98/.107/.108/.113/.132/.143/.151/.156/.159/.162/.169 — read 2026-06-26.

> Honest boundary (§11.4.6): subtasks decompose the *stated* parent-task work; no new scope is invented. All complexity figures are sizing `TARGET`s, never date commitments; device-bound subtasks (iOS HVPN-P1-122.*, Android HVPN-P1-121.*) carry the §11.4.3 SKIP-with-reason fallback where no device is reachable.